PAST PAPER NUMERICAL

FROM YEAR 1991-2025

BASIC RELATIONSHIP:

1. 10 gm of H_2SO_4 has been dissolve in excess of water to dissociate it completely into ions. Calculate: (2016)
   (i) Numbers of molecules in 10 gm of H_2SO_4 (ii) Number of positive ions
2. Find the mass and number of molecules in 180000 cm^3 of H_2S gas at S.T.P? (2012)
3. In collection of 24×10^{26} molecules of C_2H_5OH, what is the number of moles? (2011,97)
4. What is mass of 3.01×10^{20} molecules of N_2? (2007)
5. Calculate the weight in gram of 3.01×10^{20} molecules of glucose? (2004)
6. The atomic mass of Na is 23 a.m.u calculate the following: (2003)
   The mass of 4.108×10^{23} atoms of Na (ii) the number of moles of Na in 4.6 g.
7. Calculate the number of moles in 3800g of: (i) $Ca(OH)_2$ (ii) SO_4 (iii) H_2 (2003)
8. Calculate the mass in 'gm' of 3.01×10^{23} molecules of water? (2002)
9. Calculate the number of moles, the number of molecules and the volume in cm^3 0.032g of CH_4 & SO_2 at S.T.P ? (2002)
10. Calculate the number of moles and number of molecules present in 9g of $C_6H_{12}O_6$. (2002)
11. The molecular weight of NaOH is 40 a.m.u find: (1999)
    (i) Its gram molecular weight (ii) Weight of 3 molecules (iii) Weight of 60,200 molecules in gm.
12. The atomic mass of Zn is 65.4 a.m.u, Calculate (1998)
    (i) the mass of 204×10^{24} atoms if Zn in gram
    (ii) the number of moles and also calculate the number of atoms in 10.9 gm of Zn.
13. A sample contain chlorine gas at S.T.P has a volume of 800 cm^3; calculate the following: (1996)
    (i) The number of moles of chlorine (ii) the mass of the sample (iii) the number of chlorine molecules in the sample
14. How many atoms of carbon are present in 10gm of coke? (1993)
15. How many molecules in water are present in 5.4 gm of H_2O? (1992)
16. The atomic weight of gold is 197.2 a.m.u (1991)
    (i) What is the weight of one gold atom (ii) how many atoms are there in 3.29 gm of gold (iii) how many atoms are in 7.58 gm of Au ($_{79}Au^{197}$)

STOICHIOMETRY RELATIONSHIP:

1. Chlorine gas is produced on an industrial scale by electrolysis of an aqueous solution of NaCl: (2024)
   $$2NaCl + 2H_2O \rightarrow 2NaOH + H_2 + Cl_2$$
   Calculate:
   i)The volume of Cl_2 at STP, produced from 58.5g of NaCl.
   ii) The total number of moles Cl_2 produced.
2. Calculate the volume of CO_2 gas produced at 25°C and 740 torr pressure by the combustion of 20 grams of CH_4. (2022)
   $$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$$
3. If 53.5g of NH_4Cl is heated with $Ca(OH)_2$, how many grams of NH_3 is produced? Also find the volume of NH_3 at S.T.P according to the following equation: (2021)
   $$2NH_4Cl + Ca(OH)_2 \rightarrow 2NH_3 + CaCl_2 + 2H_2O$$
4. 100g of KNO_3 are heated to redness. what volume of oxygen at 39°C and 765 mm of hg pressure will evolve? (2018,2006)
   $$2KNO_3 \rightarrow 2KNO_2 + O_2$$
5. $CaCO_3 \rightarrow CO_2 + CaO$; $CaCO_3$ is often used to generate CO_2 gas in industry. If 200g of $CaCO_3$ is strongly heated, what volume of CO_2 will be obtained at 30°C and 1200 torr? (2017)
6. What volume of CO_2 measured at 20 °C and 720 torr pressure will produce by the reaction between 200gm of Na_2CO_3 and HCl. (2016)
   $$Na_2CO_3 + 2HCl \rightarrow 2NaCl + CO_2 + H_2O$$
7. Calculate the volume of nitrogen gas produce by heating 900g of ammonia of 1°C and 823 torr pressure? (2014)
   $$2NH_3 \rightarrow N_2 + 3H_2$$
8. 54 g of dinitrogen penta oxide (N_2O_5) is decomposed on heating as $N_2O_5 \rightarrow 4NO_2 + O_2$ find out the volume NO_2 and O_2 product at S.T.P? (2011,05,99)
9. Complete combustion of CH_4 give the following reaction: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ calculate the mass and volume of H_2O in 46g of CH_4 at S.T.P? (2010)
10. Zn react with H_2SO_4 (dilute) as given: $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$, calculate the mass of $ZnSO_4$, the volume of H_2 and the number of molecules of H_2 which will be produced by reacting 43.5g of Zn with H_2SO_4 at S.T.P? (2008)
11. Find the mass of $KClO_3$ required to prepare 3.2g of O_2; (2007)
    $$2KClO_3 \rightarrow 2KCl + 3O_2$$

12. 30g of lime stone ($CaCO_3$) was heated (i) write the equation of the reaction (ii) calculate the mass of CO_2 produced (iii) calculate the volume of CO_2 at S.T.P. (2004)
13. 73.5g of $KClO_3$ is decomposed on heated as following: $2KClO_3 \rightarrow 2KCl + 3O_2$; Calculate: (2003)
    (i) mass of KCl (ii) volume of O_2 at S.T.P?
14. Find the mass, number of moles, number of molecules and volume of CO_2 at S.T.P obtained by burning 6g carbon. Gives also the balanced chemical equation. (2001)
15. Complete combustion of CH_4 gives the following reaction: (2000)
    $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$, Calculate the volume of O_2 from 30 g of CH_4 at N.T.P?
16. Calculate the volume of oxygen at S.T.P that many be obtained by complete decomposition of 54.3g of $KClO_3$ on heating in the presence of MnO_7 as catalyst $2KClO_3 \rightarrow 2KCl + 3O_2$ (1995)
17. How many grams in chlorine are required to prepare 7.75 dm^3 of chloro-benzene. The equation of the reaction is: $C_6H_6 + Cl_2 \rightarrow C_6H_5Cl_3 + HCl$. (1994)
18. What amount of H_2SO_4 will be required to convert 6.45 g of Zn completely into $ZnSO_4$ and hydrogen gas according to the following equation? $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$ (1993)

LIMITING REACTANT:

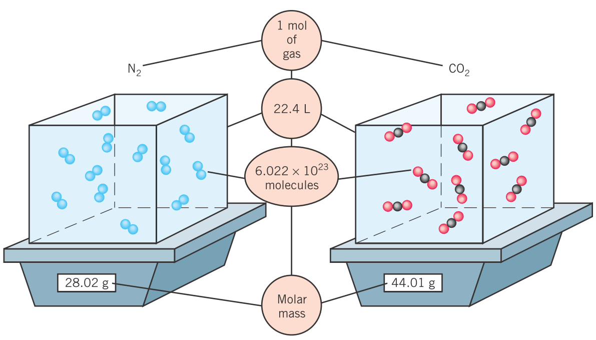
1. 53.5g of Aluminum is burnt with 22.4g of oxygen to give brilliant white flame of $2Al_2O_3$. (2024)
   $$4Al + 3O_2 \rightarrow 2Al_2O_3$$
   Which reactant is limiting reactant? Also find the mass of $2Al_2O_3$.
2. Combustion reaction of ethene (C_2H_4) in air to form CO_2 & H_2O is given in the following equation: (2023)
   $$C_2H_4 + 3O_2 \rightarrow 2CO_2 + 2H_2O$$
   If the mixture containing 2.8 gm of C_2H_4 & 6.4 gm O_2 is allowed to ignite, identify the limiting reactant and determine the mass of CO_2 gas that will be formed.
3. Find the mass of the CH_3OH produced, when 356 gm of CO is mixed with 65.0 gm of H_2. (2018)
   $$CO + H_2 \rightarrow CH_3OH$$
4. $ZnCl_2$ is prepared by the reaction: $Zn + 2HCl \rightarrow ZnCl_2 + H_2$; 6.54g of Zn reacts with 73g of HCl. Find the limiting reactant and mass of $ZnCl_2$ produced. (2014)
5. Find the weight of barium sulphate precipitated on adding a solution containing 82g of potassium sulphate to a solution 82g of barium chloride. $K_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2KCl$ (2013)
6. What is the minimum mass of Al $(OH)_3$ that can be obtained by the reaction of 13.4g of $AlCl_3$ 10g of NaOH according to the following equation? $AlCl_3 + 3NaOH \rightarrow Al(OH)_3 + 3NaCl$ (2012)
7. For the reaction of $3Mg + N_2 \rightarrow Mg_3N_2$; 1.5g of N_2 react together. What is the actual: Mg_3N_2 formed and which element is the limiting reactant? (2009)
8. You are provided with 6g of C and 100g of O_2. Calculate the amount of CO_2 prepared by reacting then. Which of them is limiting reacting? (2002)
9. At the high temperature, Sulphur combines with iron to form brow-black iron sulphide. $Fe + S \rightarrow FeS$. In one experiment 78.2g of Fe are allowed to react with 55.7g of Sulphur. Which of the two is limiting reactant? Calculate the mass of FeS obtained? (2000)

PERCENTAGE YIELD:

1. $AlCl_3$ is produced by the following reactant at 500 °C. $2Al + 3Cl_2 \rightarrow 2AlCl_3$, when 160 gm of aluminum is reacted with the excess of chlorine gas, 650 gm of $AlCl_3$ is obtained. Calculate: (a)Theoretical yield (b) % yield (2023)

PARAMETERS	STP	NTP	RTP
T	0°C=273.15K=32°F	20°C=293.15K=68°F	25 C =298.15K=80°F
P & n	01 atm and 01 mol		
v	$22.4dm^3=22400cm^3=0.02241m^3$		$24.5L = 24,500cm^3 = 0.0245m^3$

Moles of Two Different Gases At STP, one mole of any gas has the same volume, the same number of molecules, but different masses.



M. Bilal

A Stoichiometry Adventure

Fundamentals of Stoichiometry

1 For the following reactions, write the ratios that can be established among molar amounts of the various compounds:

(a) $2\ H_2 + O_2 \rightarrow 2\ H_2O$

(b) $2\ H_2O_2 \rightarrow 2\ H_2O + O_2$

(c) $P_4 + 5\ O_2 \rightarrow P_4O_{10}$

(d) $2\ KClO_3 \rightarrow 2\ KCl + 3\ O_2$

2 In an experiment carried out at very low pressure, 1.3×10^{15} molecules of H_2 are reacted with acetylene, C_2H_2, to form ethane, C_2H_6, on the surface of a catalyst. Write a balanced chemical equation for this reaction. How many molecules of acetylene are consumed?

3 ■ Sulfur, S_8, combines with oxygen at elevated temperatures to form sulfur dioxide. **(a)** Write a balanced chemical equation for this reaction. **(b)** If 200 oxygen molecules are used up in this reaction, how many sulfur molecules react? **(c)** How many sulfur dioxide molecules are formed in part (b)?

4 ■ How many moles of oxygen can be obtained by the decomposition of 7.5 mol of reactant in each of the following reactions?

(a) $2\ KClO_3 \rightarrow 2\ KCl + 3\ O_2$

(b) $2\ H_2O_2 \rightarrow 2\ H_2O + O_2$

(c) $2\ HgO \rightarrow 2\ Hg + O_2$

(d) $2\ NaNO_3 \rightarrow 2\ NaNO_2 + O_2$

(e) $KClO_4 \rightarrow KCl + 2\ O_2$

5 In petroleum refining, hydrocarbons are often manipulated by reacting them with $H_2(g)$. If hexene, C_6H_{12}, is reacted with hydrogen to form hexane, C_6H_{14}, how many moles of hydrogen are needed to react with 453 moles of hexene?

6 ■ For the following reactions, determine the value of x.

(a) $4\ C + S_8 \rightarrow 4\ CS_2$

3.2 mol S_8 yield x mol CS_2

(b) $CS_2 + 3\ O_2 \rightarrow CO_2 + 2\ SO_2$

1.8 mol CS_2 yield x mol SO_2

(c) $N_2H_4 + 3\ O_2 \rightarrow 2\ NO_2 + 2\ H_2O$

7.3 mol O_2 yield x mol NO_2

(d) $SiH_4 + 2\ O_2 \rightarrow SiO_2 + 2\ H_2O$

1.3×10^{-3} mol SiH_4 yield x mol H_2O

7 ■ Many metals react with halogens to give metal halides. For example, iron reacts with chlorine to give iron(II) chloride, $FeCl_2$.

$$Fe(s) + Cl_2(g) \rightarrow FeCl_2(s)$$

Beginning with 10.0 g iron, what mass of Cl_2, in grams, is required for complete reaction? What quantity of $FeCl_2$, in moles and in grams, is expected?

8 How many metric tons of carbon are required to reduce 7.83 metric tons of Fe_2O_3 according to the following reaction?

$$2\ Fe_2O_3 + 3\ C \rightarrow 3\ CO_2 + 4\ Fe$$

How many metric tons of iron are produced?

Limiting Reactants

1 ■ If 3.4 mol Al and 6.2 mol Fe_2O_3 are mixed, what is the limiting reactant?

$$2\ Al + Fe_2O_3 \rightarrow Al_2O_3 + 2\ Fe$$

If 8.4 moles of disilane, Si_2H_6, is combined with 15.1 moles of O_2, which is the limiting reactant?

$$2\ Si_2H_6 + 7\ O_2 \rightarrow 4\ SiO_2 + 6\ H_2O$$

2 Generally, an excess of O_2 is needed for the reaction $Sn + O_2 \rightarrow SnO_2$. What is the minimum number of moles of oxygen required to oxidize 7.3 moles of tin?

3 In the reaction of arsenic with bromine, $AsBr_5$ will form only when excess bromine is present. Write a balanced chemical equation for this reaction. Determine the minimum number of moles of bromine that are needed if 9.6 moles of arsenic is present.

4 ■ Ammonia gas can be prepared by the reaction

$$CaO(s) + 2\ NH_4Cl(s) \rightarrow 2\ NH_3(g) + H_2O(g) + CaCl_2(s)$$

If 112 g CaO reacts with 224 g NH_4Cl, how many moles of reactants and products are there when the reaction is complete?

5 When octane is combusted with inadequate oxygen, carbon monoxide may form. If 100 g of octane is burned in 200 g of O_2, are conditions conducive to forming carbon monoxide?

6 ■ The equation for one of the reactions in the process of turning iron ore into the metal is

$$Fe_2O_3(s) + 3\ CO(g) \rightarrow 2\ Fe(s) + 3\ CO_2(g)$$

If you start with 2.00 kg of each reactant, what is the maximum mass of iron you can produce?

7 Copper reacts with sulfuric acid according to the following equation:

$$2\ H_2SO_4 + Cu \rightarrow CuSO_4 + 2\ H_2O + SO_2$$

How many grams of sulfur dioxide are created by this reaction if 14.2 g of copper reacts with 18.0 g of sulfuric acid?

8 ■ One of the steps in the manufacture of nitric acid is the oxidation of ammonia shown in this equation:

$$4\ NH_3(g) + 5\ O_2(g) \rightarrow 4\ NO(g) + 6\ H_2O(g)$$

If 43.0 kg of NH_3 reacts with 35.4 kg of O_2, what mass of NO forms?

9 ■ When $Al(OH)_3$ reacts with sulfuric acid, the following reaction occurs:

$$2\ Al(OH)_3 + 3\ H_2SO_4 \rightarrow Al_2(SO_4)_3 + 6\ H_2O$$

If 1.7×10^3 g of $Al(OH)_3$ is combined with 680 g of H_2SO_4, how much aluminum sulfate can form?

10 ■ Copper reacts with nitric acid via the following equation:

$$3\ Cu(s) + 8\ HNO_3(aq) \rightarrow 3\ Cu(NO_3)_2(aq) + 2\ NO(g) + 4\ H_2O(\ell)$$

What mass of NO(g) can be formed when 10.0 g of Cu reacts with 115 g HNO_3?

1) Calculate the value of a^0 of hydrogen atom.
2) Calculate the radius of 3^{rd} orbit of hydrogen atom.
3) Calculate the value of k in the expression $k = \frac{me^4}{8h^2\varepsilon_0 2}$ [$\therefore 1J = Kg.m^2s^{-2}$]
4) Calculate the energies for n=1 for (i) $_1He^+$ (ii) $_3Li^{+2}$
5) Calculate the energy of 1st, 2^{nd} & 3^{rd} orbit of hydrogen atom?
6) Calculate the radius of 1st, 2^{nd} & 3^{rd} orbit of hydrogen atom?
7) Calculate the angular momentum of 1^{st} & 2^{nd} orbit of hydrogen atom?
8) Calculate the radius by hydrogen atom by applying Bohr's Theory. ($h = 6.625 \times 10^{-27}$ erg-sec, $m = 9.11 \times 10^{-28}$g, $e = 4.8 \times 10^{-10}$ esu).
9) Calculate the wave number of the Line in Lyman Series when an electron jumps from orbit 3 to orbit 2.
10) Calculate the wave number of spectral line of hydrogen gas when an electron jumps from n=4 to n=2. ($R_H = 109678\ cm^{-1}$)
11) How much energy is required to make an electron of H-atom to jump from n = 2 to n = 4.
12) Calculate how much energy is required in order to remove an electron of hydrogen atom. Convert this energy into frequency and in wave numbers.

ELECTRONIC CONFIGURATION:

State 'Hund's rule': Write electronic configuration of the following:

■ Cr (Z=24) ■ Fe (Z=38) ■ Cl (Z=16) (2024)

(i) Which rule is violated in the following electronic configurations?

(1)	$1s^2, 2s^3$		(2015)
(2)	$2s^2, 2p^6, 3s^2$		
(3)	$1s^2, 2s^2, 3s^2, 2p^6$		
(4)	$1s^2, 2s^2, 2p_x^{\uparrow}$		
(5)	$1s^2, 2s^{\uparrow\downarrow}$		(2011)
(6)	$1s^2, 2p^2$		
(7)	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^9, 4s^0$		
(8)	$1s^2, 2s^{\uparrow}, 2p^5$		(2008)
(9)	$1s^2, 2s^3, 2p^5$		
(10)	$1s^2, 2s^2, 2px^{\uparrow\downarrow}, 2py^{\uparrow}, 2pz^0$		(2005)
(11)	$1s^2, 2s^3$		(1998)
(12)	$1s^2, 2p^3$		
(13)	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^4\ 4s^0$		
(14)	$1s^2, 2s^3$		(1991)
(15)	$1s^2, 2s^{\uparrow}, 2px^{\uparrow}, 2py^{\uparrow}$		
(16)	$1s^2, 2s^2, 2px^{\uparrow\downarrow}, 2py^{\uparrow\downarrow}$		

(ii) Write the electronic configuration of the following:

(1) Cu^+ (Z=29)	(2) Br^- (Z= 35)	(3) Cu (Z=29)
(4) Ag (Z=47)	(5) $_{13}Al^{3+}$	(6) $_{29}Cu^{2+}$
(7) $_{17}Cl^-$	(8) Z=06	(9) Z=13
(10) Z=19	(11) Z=24	(12) Z=36
(13) Z=29	(14) $_{08}O^{-2}$	(15) F^{-1} (Z=9)

(iii) Arrange the orbitals in ascending order of their energy according to n+l rule:

(1) 3d, 4s, 4p, 5s, 6s, 5p.	(2) 4p, 4s, 3d, 5s, 5p.
(3) 4d, 7s, 4f.	(4) 3d, 4s, 4p, 5s, 6s, 5p.

GENERAL GAS EQUATION:

1) Calculate the volume occupied by 0.9 mole of oxygen gas at 5°C and 800 torr pressure? (Supp.14)
2) The density of a certain gas $1.43 g/dm^3$ at 608 torr and 27°C. Find the molecular mass of the gas? (2014)
3) Calculate the volume occupied by 80 grams of oxygen at 30°C and 800 torr pressure. (Supp.12)
4) 1.40 dm^3 volume of a gas collected at a temperature of 27°C and pressure of 900 torr was found to have a mass 2.27g. Calculate the molar mass of the gas? (2021,12,2000)
5) What will be the volume occupied by 14g of nitrogen at 20°C and 740 torr pressure? (2005)
6) What is the density of the methane gas (CH_4) at 127°C and 3.5 atmospheres? (2004)
7) A given mass of gas occupied 76 cm^3 at 16°C and 760 torr pressure. Calculate its volume at S.T.P? (2003)
8) A quantity of a gas measures 500 ml at 35°C and 600 ml of Hg pressure. What would be the volume of the gas at 45°C and 800 mm of Hg? (1999)
9) What is the volume of 2.5 moles of nitrogen gas at S.T.P? (2003)
10) 13.2g of a gas occupied a volume of 0.918 dm^3 at 25°C and 8 atm pressure. Calculate the molecular mass of the gas? (1997)
11) Calculate the number of moles of hydrogen gas present in 1.12 dm^3 volume container at 27°C and 0.1 atmospheres pressure. (1993)

GAS CONSTANT & ITS UNITS:

Units of P & V	Value at STP	Value & units of R
Metric Units (Liter-atmosphere per mole-kelvin)	P= 1 atm V= 22.4 dm3 T= 273 k	R= 0.0821 atm.dm^3/ mol.K or L.atm/mol.K
SI Units (Pascal per mole-kelvin)	P= 101325 N/m^2 or Pascals V= 0.0024 m^3 T= 273 K	R= 8.314 J/mol.K R= 1.99 Cal/mol.K (Calorie-based Units (Calorie per mole-kelvin)

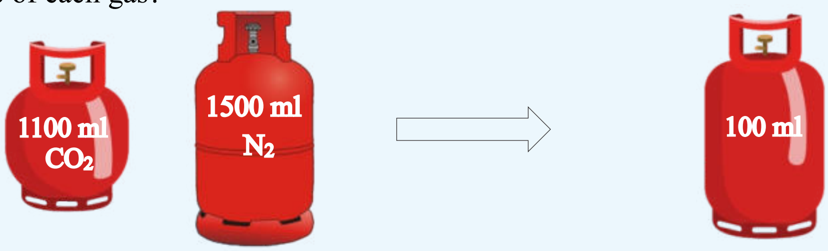
GRAHAM'S LAW OF DIFFUSION:

1) A 100 cm^3 sample of hydrogen effuses four times as rapidly as 100 cm^3 of an unknown gas. Calculate the molecular mass of the unknown gas by applying Graham's Law of Diffusion. (2024)
2) The ratio of rates of diffusion of two gases A and B is 1.5:1 respectively. If the relative (2023)
3) molecular mass of a gas A is 16, find the molecular mass of B gas.
4) If 16 ml of H_2 gas diffuse in 30 sec, what the volume of SO_2 will diffuse in the same conditions. (2016)
H_2 gas effuses from a 10 sec. Calculate the time for effuses of O_2 gas from the 20 dm^3 vessel. (2015)
5) Two gas A and B having the same effuse from a container in 20 sec respectively. If the molecular mass of gas A is 40 a.m.u. Find the molecular mass of gas B? (Supp.13)
6) At certain temperature and pressure NH_3 diffuses 1.48 time more than. HCl. If the density of NH_3 is 0.66 g/liter find the density of HCl. (2010)
7) How long would it take for oxygen to effuses from the same container at same temperature of or when helium takes 5 seconds to effuse form a hole of 10 dm^3 containers? (2009)
8) 200 cm^3 of a diffuses in the same time as 30 cm^3 of gas B. if molecular mass of gas B is 32 g/mol. Calculate the molecular mass of gas A (2003)
9) CH_4 takes 36 minutes to effuse from a container. How long will he take to effuse from the same (2002)
10) container under the same condition?
11) Calculate the molecular mass of a gas whose rate of diffuses is twice that of CH_4. (2002)
12) Compare the rates of diffusion of the following: 1) H_2 & D_2 2) CH_4 & He 3) Sf_6 & SO_2 (1999)
400 cm^3 helium gas effuse from a porous container in 20 seconds. How to long will SO_2 gas (1998)
13) take to effuse from the same container?
14) 75 cm^3 of gas A and 150 cm^3 of gas B diffuses in the same time under the same conditions of temperature and pressure. Find the molecular mass of gas A if the molecular mass B is 16g/mol. (1991)

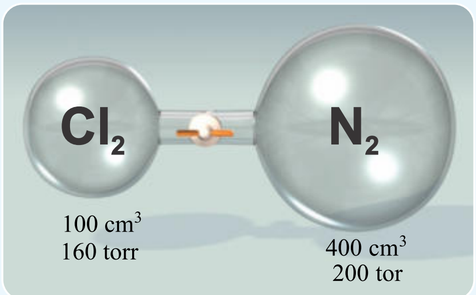
M. Bilal

DALTON'S LAW OF PARTIAL PRESSURE:

1) 40 dm^3 of hydrogen gas is collected over water at 23°C and 831 torr pressure. What would be the volume of dry gas obtained at STP (vapour pressure of water at 23°C = 21 torr). (2023)
2) 1100 ml of CO_2 at a pressure of 500 torr, 1500 ml of N_2 at a pressure of 400 torr and 800 ml of O_2 at a pressure of 600 torr are placed together in a container of 100 ml capacity. Find (2014,04,02,98,96)
3) the total pressure of each gas?



A gaseous mixture contains 0.2 mole of O_2 and 0.3 mole of CO_2 if the partial pressure of oxygen is 60 torr. Calculate the partial pressure of CO_2. (2012)
4) 1 mol of H_2 2mol of N_2 and 3 moles of O_2 are confined in a container. The total pressure of the container is 15 atm. What is the partial pressure of each gas? (2006)
5) 500 ml of oxygen gas was collected over water at 30°C C and 777.5 torr pressure calculate the mass oxygen gas? (Vapor pressure of water vapor at 50°C C is 27.5 torr) (2016)
6) At 30°C 500cm^3 of H_2 at 400 torr pressure and one dm^3 of N_2 at 600 torr pressure are transferred into a 1500 cm^3 flask. The total pressure of mixture of gases. (2015)
7) A 500 cm^3 vessel container 2g of He & 8g of CH_4. What is the total pressure of the mixture of these gases at 3°C. (supp.14)
8) A 100 cm^3 gas cylinder filled with chlorine 160 torr pressure is connected by stop-cock with another cylinder of 400 cm^3 filled with N_2 under pressure of 200 torr. What will be the total (supp.12)
9) pressure when stop-cock is opened?



A 500 cm^3 vessel contains H_2 gas at 400 torr and another 1.0 dm^3 vessel contains O_2 gas at 600 torr. If these gases are transferred to 2 dm^3 empty vessel. Calculate the total pressure of (2013)
10) the gases.
A mixture of 2.0 mol of gas A and 1.1g of another gas B (mol. Mass=44) expert a pressure of 750 torr. Calculate the partial pressure of two gases? (2008)
11) 380 cm^3 of hydrogen gas was collected over water at 23°C and 613 torr. Find the volume of dry hydrogen gas at S.T.P. ($P_{vapours}$ at 23°C is 21 torr.) (2006)
12) The volume of gas collected over water at 24°C C and 762 mm of Hg pressure is 128 ml. calculate the mass in gram of oxygen gas obtained. The pressure of water vapor at 24°C is (2002)
13) 22 mm of Hg.
A mixture of helium or hydrogen its closed in 12 dm^3 flask at 30°C,if 0.2 mole helium is present, find out of the partial pressure of each gas where has the pressure of the mixture of (1994)
14) is 2 atm
The volume of oxygen gas collected over water at 24°C and 762 mm of Hg pressure is 128ml. Calculate the mass (in g) of oxygen gas obtained. The pressure of water vapor at (1996)
15) 24°C is 22 mm of Hg.
A 5-liter vessel contain 2g of hydrogen 28g of nitrogen at 27°C. find the total pressure of t he mixture (1992)

Type #01 & 02 (Equilibrium Concentrations are Given/Required)

1) . Kc for the reaction is 55 at 700°C. Calculate the equilibrium concentration of reactants and products when the concentration of each reactant is 2.55 mole/dm^3. (2024)

$$H_2+I_2 \rightleftharpoons 2HI$$

2) 1 mole of N_2 and 1 mole of O_2 were heated at 2000°C in 10 dm^3 vessel. Calculate the equilibrium concentration of Nitric oxide, Kc for the reaction is 0.1. (2022)

$$N_2 + O_2 \rightleftharpoons 2NO$$

3) 4.6 gram of ethyl alcohol and 6.0 gram of acetic acid were kept at constant temperature until equilibrium was established 2.0 gram of acetic acid remained unused Calculate Kc. for the reaction. (2018)

$$CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$$

4) Phosphorus pentachloride decomposes in a gas phase reaction at 250°C as follows: $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$ An equilibrium, mixture in a 5.0-liter container is found to have 3.84 gm PCl_5, 9.14 gm PCl_3, and 2.84 gm of Cl_2. Evaluate Kc at 250°C. (2017)

5) For the reaction: $N_2 + 3H_2 \rightleftharpoons 2NH_3$, the equilibrium contains 0.25M nitrogen and 0.15 of hydrogen Gas at 25°C.calculate the concentration of NH_3 gas when Kc=9.6 the volume of container is 1dm^3. (2016)

6) 5.88 moles of nitrogen and 16.2 moles of oxygen are mixed and heated at 2000°C till equilibrium is established 11.28 moles of nitric oxide are formed. Calculate the value of equilibrium constant. (2014)

$$N_2 + O_2 \rightleftharpoons 2NO$$

7) 60g of acetic acid and 46g of ethyl alcohol are mixed with each other at constant temperature and are allow to attain the equilibrium.at equilibrium 58.2g of ethyl acetate and 12g of water were present. Find the equilibrium constant? (2003)

$$CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$$

8) 0.4 mol of A and 0.4 mol of B were reacted at certain temperature are allow to come to equilibrium $A+B \rightleftharpoons 2AB$. The equilibrium mixture contains 0.1 mol of AB find Kc if the volume of container was 2.0 dm^3 (2002P.E)

9) A quantity of PCl_5 is heated in a 6 dm^3 flask at 25°C till the following equilibrium was established. the equilibrium mixture contains 0.16 mol of PCl_3, 0.16 mol of Cl_2 and 0.105 mole of PCl_5.calculate the equilibrium constant? (2002P.M)

$$PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$$

10) Consider the following equation and write their equilibrium constant expressions: One mole of PCl_5 was introduced in a vessel of 10 dm^3 capacity at constant temperature. at equilibrium 0.465 moles of Cl_2 gas were present. Calculate Kc for the reaction? (2001)

11) 1.5 moles of acetic acid and 1.5 moles of ethyl alcohol are mixed with each other at constant temperature and are allow to attain the equilibrium. At equilibrium 1 mol of ethyl acetate was present. Find the equilibrium constant. (1999)

$$CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$$

12) For the reaction: $N_2 + 3H_2 \rightleftharpoons 2NH_3$, the equilibrium mixture contains 0.25M of nitrogen, 0.15M of hydrogen gas at 25°C. Calculate the concentration of NH_3 gas when Kc is 9.6.the volume of the container is dm^3. (1998)

TYPE 03: (Equilibrium concentrations are indirectly given or required by using quadratic equation or by I.C.E chart)

1) 3 moles of H_2 and 4 moles of I_2 were heated in sealed tube at given temperature at which Kc is 50. If the volume of the tube is $1dm^3$, determine the composition of equilibrium. (2013)

$$H_2 + I_2 \rightleftharpoons 2HI$$

2) Kc for the reaction: $CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$ at room temperature is 4. Calculate the equilibrium concentration of CH_3COOH when 1.66 moles of CH_3COOH and 2.17 moles of C_2H_5OH are allow to come to achieve equilibrium. (2010)
3) In the reaction: $A+B \rightleftharpoons 2C$, 7mol/dm^3 of A and 7 mol/dm^3 B were mixed and allowed to attain the equilibrium. If Kc=2.25, find out the concentration of A, B and C at equilibrium states. (2008)
4) The equilibrium constant for the reaction $N_2 + O_2 \rightleftharpoons 2NO$, at 2000°C is 0.1. Calculate the equilibrium concentration of reactants and products when initial concentration of N_2 & O_2 are 10 mol/dm^3. (2006)
5) For the reaction: $H_2 + I_2 \rightleftharpoons 2HI$, Kc is 49. Calculate the concentration of HI at equilibrium when initially one mole of H_2 is mixed with one mole of I_2 in one liter flask. (2004)
6) In a chemical reaction: $PCl_5 \rightleftharpoons PCl_3 + Cl_2$, calculate the mole of chlorine produced at equilibrium when one mole of PCl_5 is headed at 25°C in a capacity 10dm^3. (Kc=0.0410) (1992)

Type #4: (Initial Concentrations Are Required)

1) In the reaction: $H_2 + I_2 \rightleftharpoons 2HI$, when equilibrium was attained the concentration were $[H_2]$ = $[I_2]$ = [HI] =4 moles/dm3 calculate the equilibrium constant and the initial concentration of H_2 and I_2? (2015)
2) In the reaction: $A+B \rightleftharpoons 2C$ when the equilibrium is attained, the concentration of B=4 mol/dm^3 and C=6 mol/dm3. Calculate the initial and equilibrium concentration of A? (Kc=2.25) (2009)
3) When the equilibrium was attained for the reaction $A+B \rightleftharpoons 2C$, the concentration [A] = [B] was 4 mol/dm^3 and that of [C] was 6 mol/dm^3. calculate C and the initial concentration of A and B. (2003)

TYPE #05: (Miscellaneous)

1) The KC the reaction: $2HI \rightleftharpoons H_2 + I_2$ is 1.3×10^{-2} if there are 0.5 mole/dm^3 of H_2 1.5 mole/dm^3 of I_2 and 5 mole/dm^3 of HI. Predict the direction in which the reaction moves so as to achieve the equilibrium. (2019,10)

2) For the reaction: $PCl_5 \rightleftharpoons PCl_3 + Cl_2$ the equilibrium mixture at definite temperature has the partial pressure of gases as PCl_5 =0.35atm, PCl_3= 0.82atm and Cl_2=0.82atm. Calculate Kc for the reaction. (2003)

3) Write the Kc expression for the following reactions: (2006)

(i) $N_2 + 3H_2 \rightleftharpoons 2NH_3$ (ii) $CaCO_3 \rightleftharpoons CaO + CO_2$ (iii) $BiCl_3 + H_2O \rightleftharpoons BiOCl + 2HCl$

4) Write Kc expressions for the following reactions: (2004)

(i) $2SO_2 + O_2 \rightleftharpoons 2SO_3$ (ii) $3O_2 \rightleftharpoons 2O_3$ (iii) $CaCO_3 \rightleftharpoons CaO + CO_2$

5) Consider the following reactions and write their equilibrium constant expressions: (2003)

i. $3O_2 \rightleftharpoons 2O_3$ ΔH = +ve ii. $H_2 + I_2 \rightleftharpoons 2HI$ ΔH = +ve
iii. $2SO_2 + O_2 \rightleftharpoons 2SO_3$ ΔH = -ve iv. $CH_2{=}CH_2 + H_2 \rightleftharpoons CH_3{-}CH_3$ ΔH=-ve

SOLUBILITY PRODUCT

Case #01 (Solubility is given & Solubility product is required Ksp)

1) The solubility of $Mg(OH)_2$ at 25°C is 6.0×10^{-3} $g/100cm^3$. Calculate the solubility product of the $Mg(OH)_2$. (2024)
2) Write the solubility product expression of the following: (2023)
   * $PbCl_2$ * $Fe(OH)_3$ * Ag_2SO_4
3) The solubility of Mg $(OH)_2$ at 25°C 0.0046 gm/dm^3.What is the solubility product of Mg $(OH)_2$? (2022)
4) The solubility of Ca $(OH)_2$ at 25°C 0.0052 g/dm^3. What the solubility product of Ca $(OH)_2$? (supp.17)
5) The solubility of Mg $(OH)_2$ at 25°C 4.6×10^3 $gm/100cm^3$[(1)].What is the solubility product of Mg $(OH)_2$? (2016)
6) Calculate solubility product of $PbCrO_4$, when the solubility of $PbCrO_4$ is 1.0×10^{-6} g/dm^3 (2014)
7) The solubility Mg $(OH)_2$ at 25°C is 0.00764 gm/dm^3 What is its solubility product? (supp.14)
8) The solubility of AgCl at 25°C is 1.40×10^{-3} g/dm^3 its molecular mass is 143.5. Calculate the following: (2013)
   a. Molarity of AgCl solution b. Solubility product of AgCI
9) The solubility of calcium oxalate (CaC_2O_2) is 0.00764 gm/dm^3 at 25°C. Find the K_{sp} of calcium oxalate. **$CaC_2O_4 \rightleftharpoons Ca^{+2} + C_2O_2^{-2}$** (2012)
10) The given reaction is **$PbI_2 \rightleftharpoons Pb^{+2} + 2I^-$** the solubility of PbI_2 is 0.63×10^{-3} mol/dm^3 find the value of K_{sp} and express its unit. (2009)
11) What is the solubility Product for reaction: **$AgCl \rightleftharpoons Ag^+ + Cl^-$**? If he solubility of AgCl at 25°C is 1.40×10^{-3}. (2008)
12) Find the K_{sp} of $CaCO_3$ the solubility of $CaCO_3$ is 0.001 g/dm^3 at 25°C is $1.4 \times 10^{-3} g/dm^3$. **$CaCO_3, + \rightleftharpoons Ca^{+2} + CO_3^{-2}$** (2007)
13) The solubility of $BaSO_4$ is 9×10^{-3} g/dm^3 find its solubility product. (1991)

Case #02 (Solubility is required & Solubility product Ksp is given)

1) The concentration of Ag^+ ions is 1×10^4 M in AgBr Solution. Find the Minimum concentration of Br^- ions necessary to cause precipitation of AgBr. (Ksp of AgBr 4×10^{-13} M^2). (2017)
2) What is the ionic concentration of Ag and CrO_4^{-2} in a saturated solution of $AgCrO_4$, at 25°C. Ksp of $AgCrO_4$ is 13×10^{-12} mol/dm^3. (2015)
3) Calculate the solubility of AgCl at 30°C where the solubility product of AgCl is 2.5×10^{-6} mol^2/dm^6. (supp.13)
4) Calculate the solubility in $moles/dm^3$ of $PbCrO_4$, at 25°C, when its solubility product is 1.8×10^{-14} mol^2/dm^6? (supp.12)
5) The solubility product of AgCl is 9.6×10^{-11} mol/dm^3 at 25°C. Calculate solubility of AgCl in g/dm^3. (2004)
6) What is the solubility of lead iodide in mol/dm^3 at 25°C?Ksp for PbI_2 is 1×10^{-9} mol^2/dm^6. (2003)
7) The solubility product of $BaSO_4$ is 1×10^{-10} mol/dm, work out the solubility of salt in g/dm^3. (2002)

[(1)] **Conversion:** To convert grams per 100 cubic centimeters ($g/100cm^3$) to grams per cubic decimeter (g/dm^3), you need to multiply the value by 10.

Case #03 (Qsp & ppt will form or not)

1) Should AgCl precipitate from a solution prepared by mixing 400 ml of 0.1 M NaCl and 600 ml of 0.03 M of $AgNO_3$, solution? (K_{sp} of AgCl is 1.6×10^{-10} mol^2/dm^6) (2019)
2) Will Cadmium hydroxide precipitate from 0.02M solution of $CdCl_2$ at PH=10. (K_{sp} of Cd $(OH)_3$ is 2.5×10^{-14} mol^2/dm^6) (2018)
3) Should AgCl precipitate from a solution prepared by mixing 400 cm³ of 0.1 M NaCl and 600 cm³ of 0.03 M of $AgNO_3$, solution? (K_{sp} of AgCl is 1.6×10^{-10} mol/dm^3) (supp.16,2001)
4) Should $PbCrO_4$ precipitate from a solution prepared by mixing 100 cm³ of 2.5×10^{-4} M of Pb $(NO_3)_2$ and 300 cm³ of 1.5×10^{-8} Molars K_2CrO_4. (K_{sp} of $PbCrO_4$ is 1.8×10^{-24} mol^2/dm^3) (supp.15)
5) Will $PbCrO_4$, precipitate from a solution prepared by a mixing 200 cm³ of 2.5×10^{-5} M, Pb $(NO_3)_2$ and 600 cm³ of 1.5×10^{-8} M, K_2CrO_4? (K_{sp} of $PbCrO_4$ is 4.8×10^{-14}) (2005)
6) Should $PbCrO_4$ precipitate from a solution prepared by mixing 100 cm³ of 2×10^{-5} M, Pb $(CH_3COO)_2$ and 900 cm³ of 1×10^{-8} M, Na_2CrO_4. (K_{sp} of $PbCrO_4$ is 1.8×10^{-18}) (2002)

[(1)] **Numerical problem** (Previous book STBB Chemistry-11th pg.# 191):

Data:

- Concentration of $CdCl_2$: 0.02 M
- pH of the solution: 10
- Solubility product constant (*Ksp*) of $Cd(OH)_2$: 2.5×10^{-14} mol²/dm⁶.

Requirement:

We need to determine whether cadmium hydroxide ($Cd(OH)_2$) will precipitate from a 0.02 M solution of $CdCl_2$ at pH 10.

Solution:

$$Cd(OH)_2 \rightleftharpoons Cd^{2+} + 2OH^-$$
$$Ksp = [Cd^{2+}][OH^-]^2$$

1. Find the hydroxide ion concentration $[OH^-]$ from the pH:
   - pH + pOH = 14
   - pOH = 14 - pH
   - pOH = 14 - 10 = 4
   - $[OH^-]$ = -anti log (4) $=10^{-pOH} = 1\times10^{-4}$
2. Determine the initial concentration of Cd^{2+} ions:
   - The concentration of Cd^{2+} in $CdCl_2$ is the same as the concentration of $CdCl_2$, which is 0.02 M.
3. Calculate the ion product (*Qsp*) to compare with *Ksp*:

$$Qsp = [Cd^{2+}][OH^-]^2$$
$$Qsp = (0.02)(10^{-4})^2 \quad Qsp = 2 \times 10^{-10}$$

<u>If Qsp>Ksp, cadmium hydroxide will precipitate. If Qsp<Ksp, it will not precipitate.</u>

Result:

- The ion product (Qsp) for the reaction is 2×10^{-10}.
- The solubility product constant (*Ksp*) of $Cd(OH)_2$ is 2.5×10^{-14}.

Since *Qsp* (2×10^{-10}) is greater than *Ksp* (2×10^{-14}), cadmium hydroxide ($Cd(OH)_2$) will precipitate from a 0.02 M solution of $CdCl_2$ at pH 10.

M. Bilal

Case # 1 (Given: rate constant (K) and required: rate of reaction)

1) Decomposition of NO_2 into NO and O_2 is 2nd order reaction: (2023)

$$2NO_2 \rightarrow 2NO + O_2$$

If the rate constant at certain temperature is 3.8×10^{-8} $dm^3mol^{-1}sec^{-1}$ and initial concentration of NO_2 is 0.38M, calculate the initial rate of reaction.

2) For the decomposition of ethyl chloro-carbonate ($ClCO_2.C_2H_5$): (Supp.2016,2006,04, 1996)

$$ClCO_2.C_2H_5 \rightarrow CO_2 + C_2H_5Cl$$

$K=1.3\times10^{-3}$ s^{-1} at 200°C. What is the initial rate when the initial concentration of $ClCO_2.C_2H_5$ is 0.30M.

3) The rate constant for the reaction $2NO_2 \rightarrow 2NO + O_2$ is 1.8×10^{-8} $dm^3mol^{-1}sec^{-1}$ if the initial concentration of NO_2 is 2M, what is the rate of reaction? (2012,1997,94)

Case # 2 [Given: rate of reaction and Required : rate constant (K)]

1) For the decomposition of ethyl chloro-carbonate ($ClCO_2.C_2H_5$): (supp.2013)

$$ClCO_2.C_2H_5 \rightarrow CO_2 + C_2H_5Cl$$

$K=1.3\times10^{-3}$ s^{-1} at 200°C. Find the initial concentration of $ClCO_2.C_2H_5$?

2) For the reaction. $2N_2O_5 \rightarrow 4NO_2 + O_2$; calculate the rate constant when the initial concentration of N_2O_5 is 10.8 g/mol^3 while the initial rate is 2.96×10^{-4} mol/dm^3 s. (2016)

3) For the reaction $F_2 + 2ClO_2 \rightarrow FClO_2$ calculate the following: (2015)
   a. Rate expression
   b. order of reaction
   c. Rate constant when the initial concentration of F_2 is 0.1 mol/dm^3, ClO_2 is 0.01 mol/dm^3 and the reaction is 1.2×10^{-3} mole/dm^3.sec.

4) For the reaction $F_2 + 2ClO_2 \rightarrow FClO_2$.Calculate the rate constant when the initial concentration of F_2 is 0.1M, ClO_2 is 0.01M and the rate of reaction is 1.2×10^{-3} M/sec. (2008)

5) Calculate the rate constant for the reaction **A+B →AB**.when the initial concentration of both reactants is 0.1 mol/dm^3 of each and rate of reaction is 3.02×10^{-4} mol/dm^3 s: (2001)

CASE#03 [mixture of case 1 and 2]

1) The rate constant for the decomposition of NO_2; $2NO_2 \rightarrow 2NO + O_2$ is 1.8×10^{-3} dm^3 $mol^{-1}s^{-1}$. (2017, Supp.16, 1998)
   a. Write down the rate expression.
   b. Find the initial rate when concentration of NO_2 is 0.75M.
   c. Find the rate constant when the initial concentration of NO_2 is doubled.

2) Consider the following reaction and answer the following question: **A+B → C** (2010)
   a. Write rate expression with correct unit of rate of reaction
   b. Will specific rate constant increase, decrease or remain unchanged if concertation of A and B is double?

3) The rate constant for the decomposition of NO_2 in the equation $2NO_2 \rightarrow 2NO + O_2$ is 1.8×10^{-3} M/sec. what is the initial rate when the initial concentration of NO_2 is 0.75M? what is the rate constant when the initial concentration of NO_2 is double? (2009)

4) For the reaction; **A+B→AB** find the following: (2005)
   a. Rate constant if initial concertation of each reactant is 0.2M and rate of reaction is 9.04×10^{-3} M/sec.
   b. Rate constant if the initial concertation of B is double in the above given equation.

5) The decomposition of formic acid catalyzed by a strong acid. $HCOOH \rightarrow CO + H_2O$ (2003)
   a. Write its rate expression
   b. What will be the effect on this rate, if the concertation of acid is double? (2002)

6) Consider the following reaction:

a. $CH_3CHO \rightleftharpoons CH_4 + CO_2$, experimentally this reaction is found to be 2nd order. The rate at certain temperature is 0.18 $mol/dm^3.sec^{-1}$ when $[CH_3CHO] = 0.10\ mol/dm^3$

b. Write the rate expression for this reaction.

c. Determine the rate constant.

d. Calculate the rate of reaction when $[CH_3CHO]$ is 0.20 mol/dm^3.

Case # 04 [Given: table of conc. of reactant with rate of reaction and required: order of reaction]

1. Determine order of reaction from the following data: (2013)

Rate of reaction ($mole/dm^3.s$)	[A] ($mole/dm^3$)	[B] ($mole/dm^3$)
8×10^{-4}	0.1	0.1
16×10^{-4}	0.2	0.1
16×10^{-4}	0.1	0.2

2. Determine order of reaction from the following data: (2009, 2019)

Rate of reaction (M/sec)	[A] ($mole/dm^3$)	[B] ($mole/dm^3$)
1×10^{-3}	0.1	0.1
4×10^{-3}	0.2	0.1
3×10^{-3}	0.1	0.3

3. Write rate expression find the value of rate constant Determine order of reaction from the following data: (2007)

Rate of reaction (M/S)	[NO] (M)	$[O_2]$ (M)
2×10^{-3}	0.1	0.1
8×10^{-3}	0.2	0.1
6×10^{-3}	0.1	0.3

4. Determine the Order of the reaction from data given below:

$$A+B\rightarrow PRODUCT$$ (2018)

Experiment number	Rate of reaction ($mole/dm^3.s$)	[A] ($mole/dm^3$)	$[O_2]$ ($mole/dm^3$)
1.	3×10^{-3}	0.1	0.1
2.	6×10^{-3}	0.2	0.1
3.	9×10^{-3}	0.1	0.3

5. Write rate law. Determine the order of a reaction by using the following data:

$$A+B\rightarrow AB$$ (2022)

Experiment number	[A]	[B]	$\frac{dx}{dt}$
1.	0.1 M	0.1 M	$2\times10^{-3}\ Ms^{-1}$
2.	0.2 M	0.1 M	$8\times10^{-3}\ Ms^{-1}$
3.	0.1 M	0.3 M	$6\times10^{-3}\ Ms^{-1}$

Concentration Units:

1) 0.4g of NaOH is dissolved in 100g of water calculate the Molality of solution.

2) 1.5g acetic acid is dissolved in water to make 200 cm^3 of the solution. Find the concentration of solution in Molarity.
3) Determine the mass of HCl required preparing 400ml of 0.68M solution.

4) A solution is prepared by dissolving 4.9g H_2SO_4 in 250g of water. Calculate the Molality of solution.
5) What will be the Molarity of NaOH solution if 38.6g of NaOH is present in a 2.5 dm^3 of solution?
6) A 500 cm^3 solution is prepared by dissolving 1.26g of HNO_3 in water. Calculate the Molarity of solution?
7) Determine the weight of solute required to prepare 5 dm^3 of 1.2M NaOH solution.
8) Find the Molarity of 0.5g NaOH in 250 cm^3 of aqueous solution.

9) Determine the mass of HCl required to prepare 400 ml of 0.85M HCl solution.

10) 1.2 g of acetic acid (CH_3COOH) 5.4% dissociated is dissolved in water to make 200cm^3 of the solution. Find the concentration of the solution in Molarity.
11) 45 g glucose is dissolved in 72 g of water. Calculate the mole fraction of glucose & water.
12) Calculate the molality of 12% urea (urea = 60g/mol)

13) 1 g water contains 6.34×10^{-3} mg Mg^{2+} ions, calculate the concentration in ppm.

14) What is the per cent by weight of NaCl if 1.75 g of NaCl is dissolved in 5.85 g of water.
15) A sample of seawater has a chloride concentration of 1.94% (m/v). What is the chloride concentration of this seawater in ppm?
16) A sample of beer has an ethanol concentration of 4% (v/v). What is the ethanol concentration of this beer in ppm?
17) A solution has a concentration of 1.25 g. L^{-1}. What is its concentration in ppm?

18) A solution has a concentration of 0.5 mg mL^{-1}. What is its concentration in ppm?

19) 150 mL of an aqueous sodium chloride solution contains 0.0045 g NaCl. Calculate the concentration of $NaCl_{(aq)}$ in parts per million (ppm).

Raoult's Law:

1. Calculate the vapor pressure lowering caused by the addition of 100 g of sucrose (mol mass = 342) to 1000 g of water if the vapor pressure of pure water at 25°C is 23.8 mm Hg.
2. 53.94 g of a substance of molecular mass 182 is dissolved in 1000 g of water at 20°C. At this temperature the vapor pressure of water is 17.5 mm Hg. Calculate the vapor pressure of this dilute solution.
3. The vapor pressure of methyl alcohol at 298 K is 96 torr. Its mole fraction in a solution with ethyl alcohol is 0.305, what is its vapor pressure if it obeys Raoult's law.
4. An aqueous solution contains 30% by weight of a liquid A (molecular mass 120) has a vapor pressure of 160 mm at 310 K. Find the vapor pressure of pure liquid A (the vapor pressure of water at 310 K is 150 mm).

1. Calculate the heat of formation of M_2A at 25℃:(2016)
   a. $2M + \frac{1}{2}A_2 \rightarrow M_2A$ ΔH=?
   b. $M + Y_2A \rightarrow MAY + \frac{1}{2}Y_2$ ΔH=-43.2 KJ/mol
   c. $Y_2 + \frac{1}{2}A_2 \rightarrow Y_2A$ ΔH=-68.5 KJ/mol
   d. $M_2A + Y_2A \rightarrow 2MAY$ ΔH=-63.2 KJ/mol

2. Calculate the heat of formation of CH_3OH form the following data:(2015)
   a. $C + 2H_2 + \frac{1}{2}O_2 \rightarrow CH_3OH$ ΔH_f=?
   b. $C + \frac{1}{2}O_2 \rightarrow CO$ ΔH_f=-11 KJ/mol
   c. $H_2 + \frac{1}{2}O_2 \rightarrow H_2O$ ΔH_f=-268 KJ/mol
   d. $CH_3OH + O_2 \rightarrow CO + 2H_2O$ ΔH_f=-268 KJ/mol

3. Calculate the heat of formation form the following data:(2014)
   a. $4AX_3 + 5Y_2 \rightarrow 4AY + 6X_2Y$ ΔH_f=?
   b. $\frac{1}{2}A_2 + \frac{3}{2}X_2 \rightarrow AX_3$ ΔH_f=-11.0 KJ/mol
   c. $X_2 + \frac{1}{2}Y_2 \rightarrow X_2Y$ ΔH_f=-57.8 KJ/mol
   d. $\frac{1}{2}A_2 + \frac{1}{2}Y_2 \rightarrow AY$ ΔH_f=+21.6 KJ/mol

4. Calculate the heat of formation form the following data:(2013)
   a. $2Na + \frac{1}{2}O_2 \rightarrow Na_2O$ ΔH=?
   b. $Na + H_2O \rightarrow NaOH + \frac{1}{2}H_2$ ΔH=-43.2 Kcal/mol
   c. $Na_2O + H_2O \rightarrow 2NaOH$ ΔH=-63.2 Kcal/mol
   d. $H_2 + \frac{1}{2}O_2 \rightarrow H_2O$ ΔH=-68. Kcal/mol

5. Calculate the ΔH for the reaction: (2012)
   a. $3Mg + N_2 \rightarrow Mg_3N_2$ ΔH=?
   b. $3Mg + 2NH_3 \rightarrow Mg_3N_2 + 3H2$ ΔH=-371KJ/mol
   c. $\frac{1}{2}N_2 + \frac{3}{2}H_2 \rightarrow NH_3$ ΔH=-46KJ/mol

6. Calculate the heat of formation of N_2O form the following data: (2010)
   a. $C + 2H_2 + \frac{1}{2}O_2 \rightarrow CH_3OH$ ΔH=?
   b. $C + O_2 \rightarrow CO_2$ ΔH=-394KJ/mol
   c. $H_2 + \frac{1}{2}O_2 \rightarrow H_2O$ ΔH=-286KJ/mol
   d. $CH_3OH + \frac{3}{2}O_2 \rightarrow CO_2 + 2H_2O$ ΔH=-726KJ/mol

7. Calculate the heat formation of propane from the following data:(2009)
   a. $3C + 4H_2 \rightarrow C_3H_8$ ΔH=?
   b. $C + O_2 \rightarrow CO_2$ ΔH=-394KJ/mol
   c. $H_2 + \frac{1}{2}O_2 \rightarrow H_2O$ ΔH=-286KJ/mol
   d. $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$ ΔH=-2200KJ/mol

8. Calculate the heat formation of N_2O form the following data: (2008)
   a. $2NO_2 \rightarrow N_2O_4$ ΔH=?
   b. $\frac{1}{2}N_2 + 2O_2 \rightarrow NO_2$ ΔH=+33.95KJ/mol
   c. $N_2 + 2O_2 \rightarrow N_2O_4$ ΔH=+9.3KJ/mol

9. Calculate the heat formation of ethane at 25℃ form the following data: (2007,2018)
   a. $2C + 3H_2 \rightarrow C_2H_6$ ΔH=?
   b. $C + O_2 \rightarrow CO_2$ ΔH=-394 KJ/mol
   c. $H_2 + \frac{1}{2}O_2 \rightarrow H_2O$ ΔH=-286 KJ/mol
   d. $C_2H_6 + \frac{7}{2}O_2 \rightarrow 2CO_2 + 3H_2O$ ΔH=-1560 KJ/mol

10. Calculate the heat formation of form the following data: (2006)
   a. $2C + 3H_2 + \frac{1}{2}O_2 \rightarrow C_2H_5OH$ ΔH=?
   b. $C + O_2 \rightarrow CO_2$ ΔH=-394 KJ/mol
   c. $H_2 + \frac{1}{2}O_2 \rightarrow H_2O$ ΔH=-286 KJ/mol
   d. $C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$ ΔH=-1396 KJ/mol

11. Find the heat of formation of acetic acid 25℃ : (2005)
   a. $2C + 2H_2 + O_2 \rightarrow CH_3OOH$ ΔH=?
   b. $C + O_2 \rightarrow CO_2$ ΔH=-394KJ/mol
   c. $H_2 + \frac{1}{2}O_2 \rightarrow H_2O$ ΔH=-296KJ/mol
   d. $CH_3OOH + 2O_2 \rightarrow 2CO_2 + 2H_2O$ ΔH=-890.34KJ/mol

12. Determine ΔH for the reaction: (2003P.M)
   a. $2Na + H_2O\ \frac{1}{2}O_2 \rightarrow 2NaOH$ ΔH=?
   b. $Na + \frac{1}{4}O_2 \rightarrow \frac{1}{2}Na_2O$ ΔH=-50Kcal
   c. $Na_2O + H_2O \rightarrow 2NaOH$ ΔH=-56Kcal

13. Calculate the heat of formation of methane from the following data: (2003 P.E)
   a. $C + H_2 \rightarrow CH_4$ ΔH=?
   b. $C + O_2 \rightarrow CO_2$ ΔH=-394KJ/mol
   c. $H_2 + \frac{1}{2}O_2 \rightarrow H_2O$ ΔH=-286KJ/mol
   d. $CH_4 + O_2 \rightarrow CO_2 + 2H_2O$ ΔH=-890.34KJ/mol

14. Calculate the heat of formation of Acetylene (C_2H_2) form the following data: (2002)
   a. $2C + H_2 \rightarrow C_2H_2$ ΔH=?
   b. $C + O_2 \rightarrow CO_2$ ΔH=-394KJ/mol
   c. $C_2H_2 + \frac{5}{2}O_2 \rightarrow 2CO_2 + H_2O$ ΔH=-1200KJ/mol
   d. $H_2 + \frac{1}{2}O_2 \rightarrow H_2O$ ΔH=-286KJ/mol

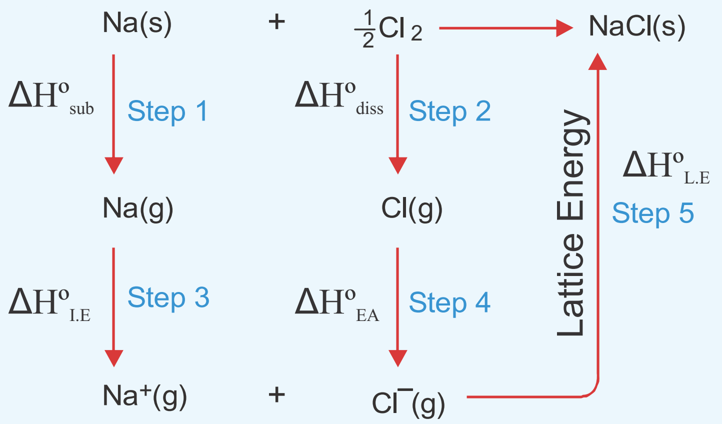
15. Calculate the heat formation of Benzene at 25℃: (1997,2017)
   a. $6C + 3H_2 \rightarrow C_6H_6$ ΔH=?
   b. $C + O_2 \rightarrow CO_2$ ΔH=-394KJ/mol
   c. $H_2 + \frac{1}{2}O_2 \rightarrow H_2O$ ΔH=-286KJ/mol
   d. $C_6H_6 + O_2 \rightarrow CO_2 + H_2O$ ΔH=-3267KJ/mol

16. Calculate the heat of formation of Fe_2O_3 t 25℃ : (2018,2024)
   a. $2Fe + \frac{3}{2}O_2 \rightarrow Fe_2O_3$ ΔH_f =?
   b. $H_2 + \frac{1}{2}O_2 \rightarrow H_2O$ ΔH = -286 KJ/mol
   c. $Fe_2O_3 + 3H_2O \rightarrow 2Fe(OH)_3$ ΔH=+289 KJ/mol
   d. $Fe + 3H_2O \rightarrow Fe(OH)_3 + \frac{3}{2}H_2$ ΔH=+161 KJ/mol

17. Calculate ΔH_f of the following data: (2021,2022) a. $6C + 3H_2 \to C_6H_6$ $\Delta H = ?$ b. $C + O_2 \to CO_2$ $\Delta H = -394$ KJ/mol c. $H_2 + ½$ ② $\to H_2O$ $\Delta H = -286$ KJ/mol d. $C_6H_6 + — O_2 \to CO_2 + H_2O$ $\Delta H = -3267$ KJ/mol	18. Calculate the standard enthalpy of formation ΔH_f of ($C_3H_8O_3$) from the data given below: (2023) a. $3C + 4H_2 + \frac{7}{2} O_2 \to C_3H_8O_3$ $\Delta H_f = ?$ b. $C + O_2 \to CO_2$ $\Delta H = -393.5$ KJ/mol c. $H_2 + ½$ ② $\to H_2O$ $\Delta H = -285.8$ KJ/mol d. $C_3H_8O_3 + 3\frac{1}{2} O_2 \to 3CO_2 + 4H_2O$ $\Delta H = -2200$ KJ/mol

HABER-BORN CYCLE:

1. Determine the lattice energy for CsF (s) writing all the involved thermochemical equations stepwise, using the following data: (all the values are in KJ/mole). (2023)
   * ΔHf of CsF = -553.5
   * $\Delta H_{(E.A)}$ of F = -328.2
   * $\Delta H_{(diss)}$ of F_2 = 157
   * $\Delta H_{(sub)}$ of Cs = 76.5
   * $\Delta H_{(I.E)}$ of Cs = 375.5



A Born-Haber cycle for the formation of NaCl crystal from its elements.

قَالَ ابْنُ عَبْدِ البَرِّ: صَحَّ عَنْ أَبِي الدَّرْدَاءِ أَنَّ: لاَ أَدْرِي (نِصْفُ العِلْمِ)

(سير أعلام النبلاء –الطبقة السابعة، (١٠) مالك بن أنس بن مالك المدني: ٧٧/ 08.)

ابنِ عبد البر رضی اللہ عنہ فرماتے ہیں مجھے ابو درداء کی یہ بات بالکل درست پہنچی ہے کہ کسی ایسے سوال پر جسکا علم نہیں اپنی لاعلمی کا اظہار کر دینا نصف علم ہے۔

M. Bilal

Balancing a equation by ION-ELECTRON METHOD half reaction method:

1. Write the given equation in ionic form.
2. Identify atoms undergoing oxidation (oxidation number ↑)and reduction (oxidation number ↓).
3. Break the equation into two halves; i.e. reduction half and oxidation half.
4. Balance the half equation:
   a) Balance all other atoms except oxygen and hydrogen.
   b) Balance oxygen by adding H_2O to the side deficient in Oxygen atoms.
   c) Balance hydrogen by adding H^+ ions to the side deficient in hydrogen atoms.
   d) Balance charge by adding electrons to the side where +charge is greater.
5. Add the two halves such that electrons in two halves get cancel.
6. In **Basic Medium** add as many OH^- ions on both sides as no. of H^+ ions on one side.

Acidic medium

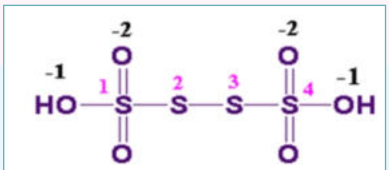
1) $Cr_2O_7^{-2} + Cl^- \rightarrow Cr^{+3} + Cl_2$ (2023)
2) $Br^{-1} + NO_3^- + H^+ \rightarrow NO + Br_2 + H_2O$ (2022, supp.2017)
3) $HNO_3 + H_2S \rightarrow NO + S + H_2O$ (2018, supp.2015, supp.14, 2014,1999)
4) $Cr_2O_7^{-2} + I_2 \rightarrow Cr^{+3} + IO_3^-$ (2024,S.2023,2021,17,13,07,06,03,02,00)
5) $Cr_2O_7^{-2} + Fe^{+2} \rightarrow Cr^{+3} + Fe^{+3}$ (2015)
6) $MnO_4 + Fe^{+2} \rightarrow Mn^{+2} + Fe^{+3}$ (supp.12,2009,2008)
7) $MnO_4^{-1} + C_2O_4^{-2} \rightarrow Mn^{+2} + CO_2$ (Supp.2023,2012)
8) $MnO_4^{-1} + Cl^{-1} \rightarrow Mn^{+2} + Cl_2$ (2019, supp.14,2010,1997)
9) $MnO_4^{-1} + NO_2^{-2} \rightarrow Mn^{+2} + NO_3^{1-}$ (1998)

1) $MnO_4^{-1} + SO_3^{-2} \rightarrow Mn^{+2} + SO_4^{-2}$ (2023,2014,2013)
2) $CrO_4^{-2} + I^- \rightarrow Cr^{+3} + IO_3^-$ (Supp. 2015)
3) $Br_2 + OH^- \rightarrow Br^- + BrO_3^- + H_2O$ (2022)
4) $MnO_4^{-1} + SO_3^{2-} + OH^- \rightarrow Mn^{+2} + SO_4^{2-}$ (2021,19,2018,11,10,06,03,02,99,98,97)
5) $Cr(OH)_3 + SO_4^{-2} \rightarrow CrO_4^{-2} + SO_3^{-2}$ (2024, S.23,17,2017, S.12,12,02,2000)
6) $Cl_2 + OH^- \rightarrow Cl^- + ClO_3 + H_2O$ (S.2015)

Basic medium

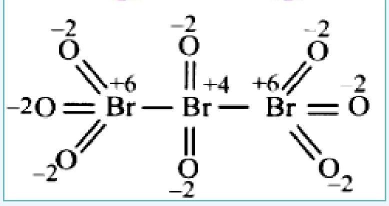
1) Cr in $Cr_2O_7^{-2}$
2) S in $H_2S_2O_7$
3) Mn in MnO_4^{-1}
4) P in H_3PO_4
5) N in NCl_3
6) Cr in CrO_7^{-2}
7) P in $P_2O_7^{-4}$
8) H in H_2
9) N in HNO_3
10) C in $C_2O_4^{-2}$
11) S in NaS_2O_3
12) Cr in $K_2Cr_2O_7$
13) S in $Na_2S_4O_6$
14) Cl in Mg $(ClO_4)_2$
15) Fe in $Fe_2(SO_4)_3$
16) C in C_2H_6O
17) Mn in $KMnO_4$
18) N in NH_4^+

a) S in $H_2S_4O_6$[1] ⟶



[1] disulfuric acid or tetra thionic acid.

b) Fe in Fe_2O_3 and Fe_3O_4[2]
c) Pb in Pb_3O_4[3]
d) Br in Br_3O_8[4] ⟶



The rules for oxidation states: (by formula)

- Each atom in a formula has an oxidation state.
- The oxidation state of Alkali metals (G-IA) is always '+1'
- The oxidation state of Alkaline earth metals (G-IIA) is always '+2'
- The oxidation state of **Fluorine** is always '-1'
- Any atom in elemental state has oxidation number '0'. For e.g., Cl in Cl_2, Br in Br_2, S in S_8, O in O_2.

OXIDATION STATE OF OXYGEN

-2 in 99% cases	'-1' in peroxides (H_2O_2, Na_2O_2)	'-1/2' in superoxide (KO_2, NaO_2)	'+1' (H^{+1}–O^{-1}–O^{+1}–F^{-1})	'+2' (OF_2)

- Oxidation state of halogens (G-VIIA) is generally '-1' because they show variable oxidation states. [Cl, Br, I, F).

JUST LOOK AT CHLORINE: it shows variable oxidation states[5] (-1 → +7) = -1,0, +1, +2, +3, +4, +5, +6, +7.

FORMULA:

✓ **Sum of all oxidation numbers/states of all atoms present in a specie = total charge on specie**

S in SO_4^{-2}

$x+4(-2) = -2$

$x - 8 = -2$

$x = -2 + 8$

$\boxed{x = +6}$

BALANCING THE EQUATION BY OXIDATION NUMBER METHOD:

1. Identify atoms which undergo oxidation and reduction.
2. After balancing atoms undergoing oxidation and reduction then balance charge by cross multiplication.
3. Balance all other atoms except oxygen and hydrogen.
4. To balance oxygen, add H_2O.
5. To balance Hydrogen atoms, add H^+ ions in **Acidic Medium**.
6. In **Basic Medium** add as many OH^- ions on both sides as no. of H^+ ions on one side.

[2] iron (II, III) oxide or iron (II, III) tetroxide. It is commonly known as magnetite.
[3] lead (II, IV) oxide or lead tetra oxide.
[4] tribromine octoxide.
[5] P-block or D-block elements generally shows variable oxidation states; such as N (-3↔+5), S (-2↔+6), and Cl (-1↔+7)